

Mohamed Sakr Structural Engineer · BIM & Digital Delivery

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Structural engineer with 10 years of experience on commercial, industrial, and residential projects across Estonia, Egypt, and Saudi Arabia — including buildings up to 15,000 m² taken through Tekla Structures from concept model to issued shop drawings. Currently at ESPLAN in Tallinn, where alongside design work I have built custom parametric Tekla components and automation rules that cut repetitive modelling time by up to 30%. Holds an MSc from TalTech, with a graduation project completed at Aalto University. Bringing structural expertise and Tekla automation experience to BIM management and construction technology roles.

CORE SKILLS

BIM & Modelling: Tekla Structure | Revit Structure | AutoCAD | Shop drawing | coordination | Clash detection

Structural Tools: FEM design | Robot Structural | ETABS | SAP2000 | SAFE | STAAD.Pro | ABAQUS

Design Codes: Eurocode | ACI-318 | ASCE-7 | ECP-203 | ECP-201 | UBC/IBC

Project Skills: Structural coordination | Quality management | BOQ preparation | Technical reporting

Digital Development: Tekla automation rules | Custom parametric components | Scripting basics

PROFESSIONAL EXPERIENCE

Structural Engineer - BIM & Modelling, ESPLAN 07/2018 – Present | Tallinn, Estonia

Structural design and Tekla Structures modelling for residential towers, industrial facilities, commercial buildings, and public infrastructure, projects ranging from 2,000 to 15,000 m². Covering the full workflow from structural layout through coordinated, issued shop drawings.

- Delivered 3D structural models and shop drawings in Tekla Structures for **Rocca towers** (multi-storey residential + parking, ~5,000 m²), **Fortaco Factory** (industrial ~12,000m²), **Lidl supermarket** (commercial ~3,000m²), and **Vindi kindergarten** (Public Use ~4,000m²)
- Led structural model coordination with architectural and MEP disciplines, catching and resolving clashes at design stage to prevent on-site conflicts and drawing revisions
- Owned quality review of structural deliverables against Eurocode requirements before issue, reducing the number of revision cycles on coordinated drawing sets
- Built parametric custom components and automation rules in Tekla Structures that cut model setup and drawing generation time by up to 30% across residential and industrial projects, tools adopted across the team as part of standard workflow

Structural Engineer, Jawda Consultants 02/2017 – 06/2018 | Cairo, Egypt

Structural analysis and RC design for high-voltage power substation structures commissioned by Saudi Electricity Co. / National Grid KSA, delivered to ACI-318, ASCE-7, and UBC/IBC codes. **Four substation projects (110/13.8 kV and 132/33 kV) in Jeddah and Makkah.**

- Completed structural analysis and RC design using STAAD.Pro across four substations; produced all calculation packages, drawings, and code compliance documentation for contractor review and approval
- Developed 3D structural models and coordinated with electrical, architectural, and mechanical disciplines to resolve structural-MEP clashes at design stage across all four facilities

Structural Engineer & site Engineer, Alhoda Consultant 2014 – 2017 | Cairo, Egypt

- RC analysis and design for residential towers (G+11), wastewater treatment plants, and steel structures using ETABS, SAP2000, and SAFE to ECP-203
- Site role covered contractor supervision and quality control on residential builds

EDUCATION

- Professional Development Programme, *kood/Jõhvi*** 2024 – 2026 | Estonia
- Scripting, process automation, and digital workflow tooling
- Master of Structural Engineering, *TalTech University*** 2020 – 2022 | Estonia
- Graduation project completed at Aalto University:
"Numerical studies of composite beam behaviour under fire"
- Bachelor of Civil Engineering, *Zagazig University*** 2009 – 2014 | Egypt

CERTIFICATIONS & REGISTRATIONS

- Disciplined Agile Scrum Master (DASM) · Project Management Institute (PMI) | 2023
- Civil Engineer in Buildings & Structures, ESTQF Level 7 · Eesti Ehitusinseneride Liit | 2024

LANGUAGES

Arabic: Native • **English:** C1 Professional • **Estonian:** B1 Intermediate